CivicProof GIS: Budget & Implementation Roadmap

Phased Deployment Schedule, Resource Allocation & Risk Matrix

1. Phased Implementation Roadmap

• Phase 1: GIS Survey & Infrastructure Tagging (Month 1 - 3)
  - High-precision GPS mapping of 1,200+ bathing ghat zones, sanitation units, RO kiosks, and medical camps in Rajahmundry.
  - Database seeding and Telugu localization of all civic asset names and landmarks.

• Phase 2: QR Placard Fabrication & Pilot Deployment (Month 4 - 6)
  - Production of weatherproof aluminium composite QR labels with reflective UV coating for outdoor riverbank durability.
  - Municipal pilot deployment with 50 sanitary inspectors testing offline draft recording.

• Phase 3: Stress Testing, CDN Edge Caching & City Poster Blitz (Month 7 - 9)
  - Simulated 250,000 concurrent user load test on routing and QR resolution APIs.
  - Installation of 500+ A2-sized City Discovery Posters at railway stations, bus terminals, and highway entry tollgates.

• Phase 4: Pushkaram Live Command Center Operations (Event Month)
  - 24/7 technical monitoring team co-located with RMC Central Police Control Room.
  - Real-time dispatching based on the Attention Queue defect dashboard.

2. Comprehensive Budget Estimate

Below is the itemized budget for turnkey municipal execution across Rajahmundry Municipal Corporation (RMC):

Component	Description	Est. Cost (INR)	Est. Cost (USD)
Physical QR Signage	2,500 Heavy-duty anodized aluminium QR placards & 500 Posters	₹8,50,000	$10,200
Cloud & Edge CDN	High-availability Kubernetes cluster, Cloudflare CDN, Tile Cache	₹9,80,000	$11,800
Inspector Hardware	120 Rugged IP68 Android tablets for sanitary and medical inspectors	₹18,00,000	$21,600
Field Survey & Training	Ground GIS survey team, inspector training workshops, manuals	₹5,20,000	$6,250
Command Center & Contingency	24/7 Operations monitoring room, emergency tech reserve	₹7,00,000	$8,400
Total Project Investment	**Full 12-Month Lifecycle including Event Execution**	**₹48,50,000**	**$58,250**

3. Risk Management & Mitigation

• Network Blackouts: Full PWA caching ensures all wayfinding and cached facilities remain accessible without mobile signal.
• QR Placard Damage: High-durability aluminium plates with manual code fallbacks printed prominently beside each code.
• High Traffic Spikes: Stateless CDN caching of public GIS tiles and static route geometries prevents server load choke.